Analysis Report

Executive summary

For the market intelligence project we collected data from three different sources to have wide insights:

1. Database: library.db which have info about books, members, borrowings and fines
2. Github API: searching for python-book-related repositories and collecting information such as star count and programming languages used.
3. Web Scraping: by scraping two categories “Travel” , “Mystery” and others and collecting info about books prices and ratings

Data collection statistics

Source	Number of records	Total attempts	Successful attempts
Database	-	1	1
Github API	25	1	1
Scarped	160	160	160

Logs

```
2026-03-09 03:40:57,168 - __main__ - INFO - Pipeline initialized
2026-03-09 03:40:57,168 - __main__ - INFO - Collecting from database:
library.db
2026-03-09 03:40:57,168 - __main__ - INFO - Collected 3 records from
database
2026-03-09 03:40:57,246 - __main__ - INFO - Collecting from API:
https://api.github.com/search/repositories
2026-03-09 03:41:01,021 - __main__ - INFO - Rate Limit: 4/10, resets at
2026-03-09 03:41:40
2026-03-09 03:41:02,677 - __main__ - INFO - Rate Limit: 3/10, resets at
2026-03-09 03:41:40
2026-03-09 03:41:04,384 - __main__ - INFO - Rate Limit: 2/10, resets at
2026-03-09 03:41:40
```

```
2026-03-09 03:41:06,055 - __main__ - INFO - Rate Limit: 1/10, resets at
2026-03-09 03:41:40
2026-03-09 03:41:07,850 - __main__ - INFO - Rate Limit: 0/10, resets at
2026-03-09 03:41:40
2026-03-09 03:41:07,850 - __main__ - WARNING - Rate limit reached.
Sleeping 33s...
2026-03-09 03:41:41,946 - __main__ - INFO - Collected 25 records
2026-03-09 03:41:42,004 - __main__ - INFO - Scraping categories:
['Travel', 'Mystery', 'Historical Fiction', 'Science Fiction', 'Fantasy',
'Romance']
2026-03-09 03:42:26,127 - __main__ - INFO - Scraped 160 books from
categories: ['Travel', 'Mystery', 'Historical Fiction', 'Science Fiction',
'Fantasy', 'Romance']
2026-03-09 03:42:26,150 - __main__ - INFO - Data exported to exports/
2026-03-09 03:42:26,157 - __main__ - INFO - Pipeline closed
```

Market insights

Popular genres

From the data collected from the database, we found that the Fiction genre had the largest distribution with 80% of the books, while Biography represented only 20%.
From the scraped data from (mystery, romance, fiction, travel, science fiction, historical fiction), we found that fantasy, romance and mystery were the most popular among these categories with percentages of 25%, 21.9% and 20%

Price trends

We found that the average price fluctuated relative to the average rating. Based on scraped data from the previously mentioned genres, the average price varied from 33 to 38 pounds, with books with average rating of 3 having the lowest average price.

Rating patterns

When comparing the average ratings of the scraped categories, Travel and Mystery, both categories showed very similar average ratings of approximately 2.7~2.9, so there was no strong difference.

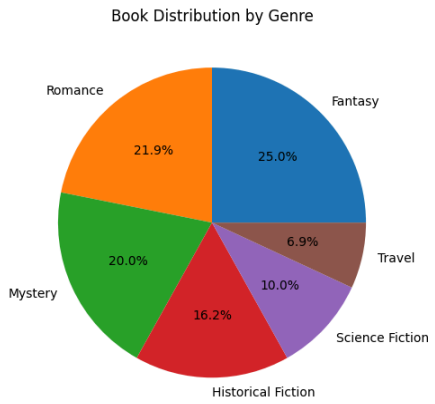
Technology trends (from GitHub)

For python-book-related repositories, the most starred repositories were “Introduction to ML with Python” and “One Python Craftsman.”

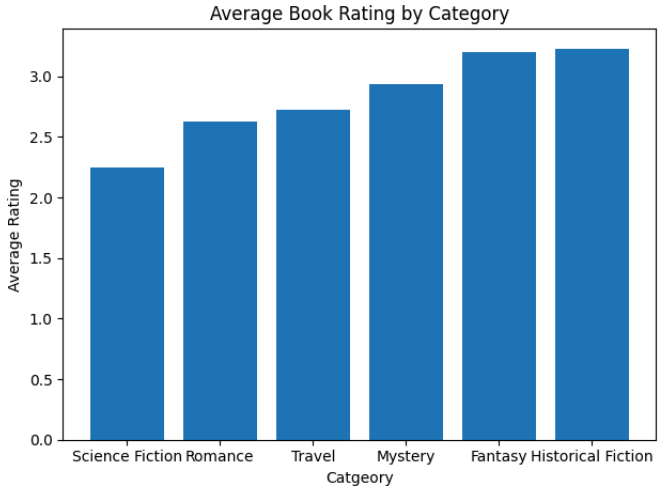
We also found that the most commonly used technology was Jupyter Notebook (48%), while Batchfile was the least used at 4%. Additionally, about 24% of repositories had unknown or unspecified languages.

Visualizations

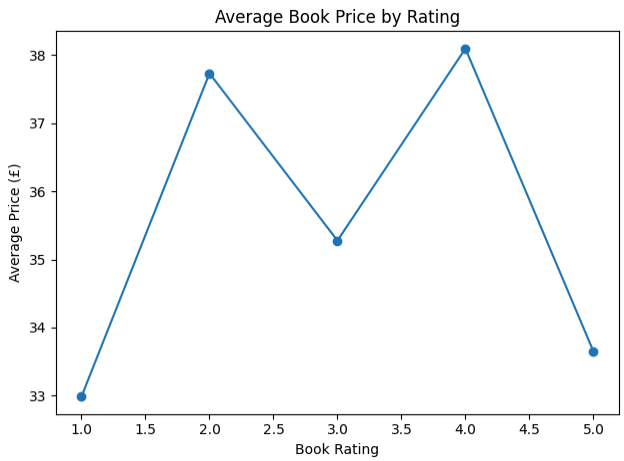
- Book distribution based on category from scraped books



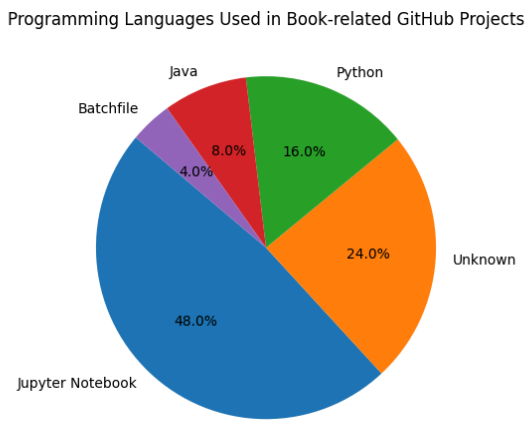
- Average book rating by category based on scraped data



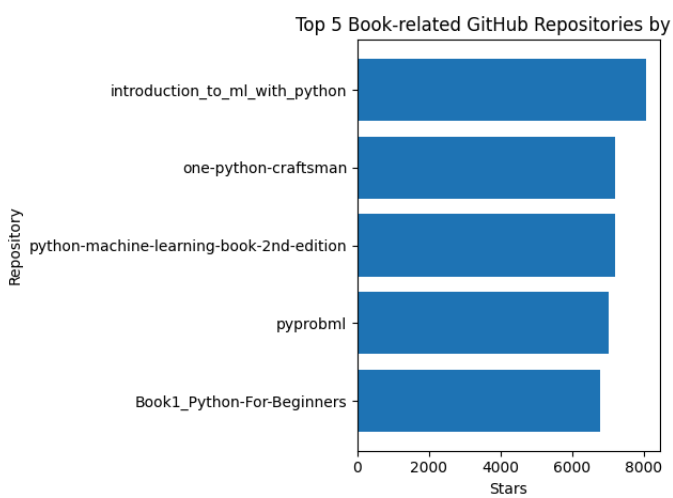
- The relationship between average book price and rating based on scraped data from the categories travel, mystery, romance, science fiction, historical fiction and fantasy



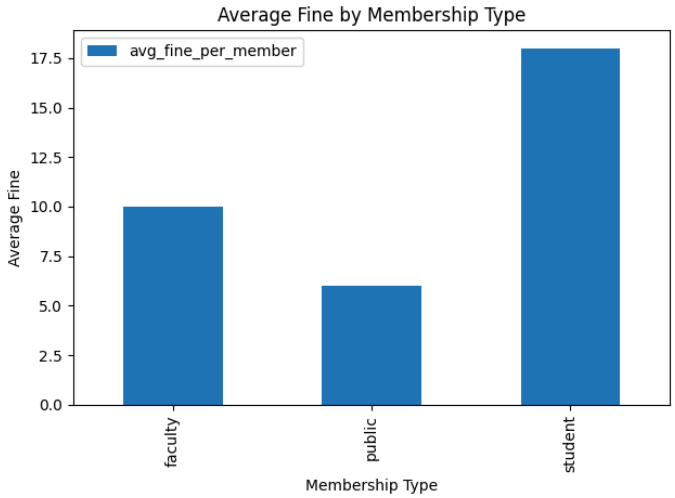
- The most commonly used programming languages based on api collected data



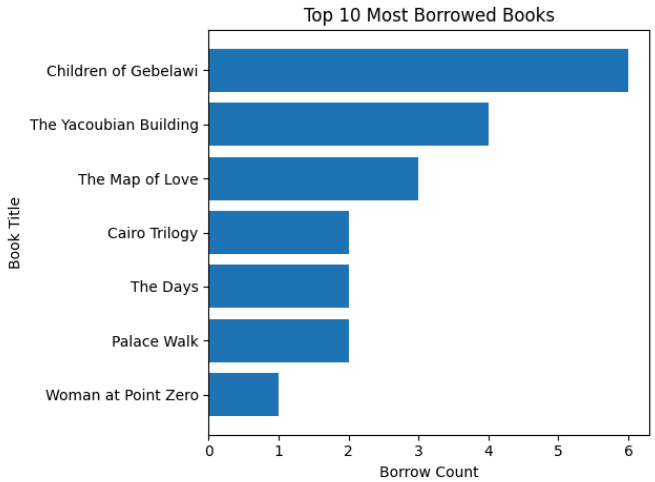
- Top rated books from github api



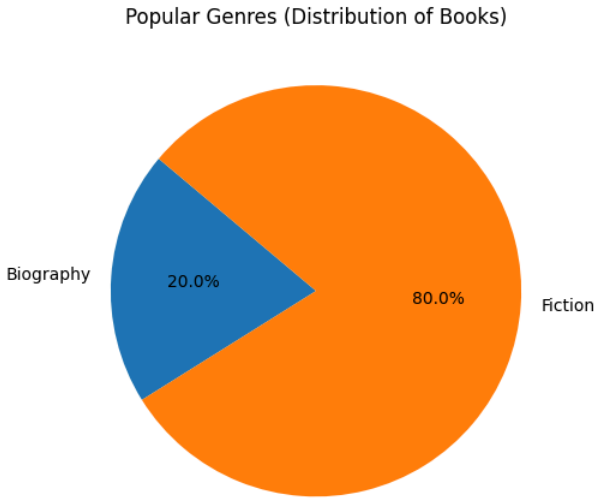
- Average fine by membership type based on [library.db]() data



- Top borrowed books based on [library.db]() data



- Popular genres based on library.db



Recommendations

Based on the collected data and analysis, several insights can help guide decisions:

- Since Fiction dominates the library collection, the library may need to expand to new genres like “Historical Fiction”, “Fantasy” and “Mystery” since they are most popular
- Books with higher prices appear to have better ratings, which may indicate higher perceived quality. Libraries or bookstores could prioritize acquiring better reviewed books even if they are slightly more expensive.
- From the GitHub analysis, Jupyter Notebook is widely used for book-related Python repositories which suggests that it’s better to use notebooks than normal python scripts